



Geometric and Negative Binomial Distributions · Mark Scheme

Further Statistics 1 Chapter 3: Geometric and Negative Binomial Distributions. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – The geometric distribution

- 1 0.3×0.7^3 M1; = 0.1029 A1.
- 2 (a) $(\frac{5}{6})^2 \times \frac{1}{6} = 0.1157$ B1. (b) $P(X > 4) = (\frac{5}{6})^4$ M1; = 0.4823 A1.
- 3 (a) $1 - 0.8^5 = 0.6723$ B1. (b) $P(X > 2) - P(X > 6) = 0.8^2 - 0.8^6$ M1; = 0.3779 A1.
- 4 (a) $p(1-p) = 0.21 \Rightarrow p^2 - p + 0.21 = 0$ M1; $p = 0.3$ (reject 0.7 as $p < 0.5$) A1. (b) $P(X > 3) = 0.7^3$ M1; = 0.343 A1.
- 5(a) $X \sim \text{Geo}(0.35)$ B1.
- 5(b) $0.65^2 \times 0.35$ M1; = 0.1479 A1.
- 5(c) $P(X > 5) = 0.65^5$ M1; = 0.1160 A1.
- 5(d) $1 - 0.65^n > 0.95 \Rightarrow 0.65^n < 0.05$ M1; $n > \frac{\ln 0.05}{\ln 0.65} = 6.95$ A1; $n = 7$ A1.

Section B – Mean, variance and memoryless property

- 6 $E(X) = \frac{1}{0.25} = 4$ B1; $\text{Var}(X) = \frac{0.75}{0.25^2} = 12$ B1.
- 7 $\frac{1}{p} = 5 \Rightarrow p = 0.2$ B1; $\text{Var}(X) = \frac{0.8}{0.2^2} = 20$ M1; standard deviation = $\sqrt{20} = 4.47$ A1.
- 8 $\frac{P(X > 7)}{P(X > 4)} = \frac{0.6^7}{0.6^4}$ M1 M1; = $0.6^3 = 0.216$ A1.
- 9 (a) $\frac{1-p}{p^2} = 6 \Rightarrow 6p^2 + p - 1 = 0$ M1; $(3p-1)(2p+1) = 0$, $p = \frac{1}{3}$ A1. (b) $E(X) = 3$ B1; $P(X = 3) = \frac{1}{3} \times (\frac{2}{3})^2 = \frac{4}{27} = 0.1481$ A1.
- 10(a) Mean = $\frac{1}{0.08} = 12.5$ B1; standard deviation = $\sqrt{\frac{0.92}{0.08^2}} = \sqrt{143.75} = 12.0$ B1.
- 10(b) $P(X > 10) = 0.92^{10}$ M1; = 0.4344 A1.
- 10(c) $P(X > 16 | X > 6) = P(X > 10)$ M1; = 0.4344 A1; the memoryless property (lack of memory) B1.
- 10(d) Mean = $3 \times 12.5 = £37.50$ B1; standard deviation = $3 \times 11.99 = £35.97$ B1.

Section C – The negative binomial distribution

- 11 $\binom{4}{2}(0.4)^3(0.6)^2$ M1; = 0.1382 A1.
- 12 $E(X) = \frac{4}{0.5} = 8$ and $\text{Var}(X) = \frac{4(0.5)}{0.5^2} = 8$ B1; $P(X = 6) = \binom{5}{3}(0.5)^6$ M1; = 0.1563 A1.
- 13 $X \sim \text{NB}(2, \frac{1}{6})$ B1; $\binom{4}{1}(\frac{1}{6})^2(\frac{5}{6})^3$ M1; = 0.0643 A1.
- 14 (a) $\binom{4}{2}(0.7)^3(0.3)^2$ M1; = 0.1852 A1. (b) $P(X = 3) + P(X = 4) = 0.7^3 + \binom{3}{2}(0.7)^3(0.3)$ M1; = 0.6517 A1.
- 15(a) $\binom{9}{2}(0.2)^3(0.8)^7$ M1; = 0.0604 A1.
- 15(b) $E(X) = \frac{3}{0.2} = 15$ B1; $\text{Var}(X) = \frac{3(0.8)}{0.2^2} = 60$ B1.
- 15(c) $1 - [P(X = 3) + P(X = 4) + P(X = 5)]$ M1; = $1 - (0.008 + 0.0192 + 0.03072)$ A1; = 0.9421 A1.
Alternative: fewer than 3 sales in 5 calls, $P(Y \leq 2)$ for $Y \sim \text{B}(5, 0.2)$.
- 15(d) She needs 2 more sales, the second on the 5th further call: $\binom{4}{1}(0.2)^2(0.8)^3$ M1; = 0.0819 A1.



Roar-level challenge

16 (a) A wins if the first six is on roll 1, 3, 5, ...: $\frac{1}{6} + \left(\frac{5}{6}\right)^2 \frac{1}{6} + \left(\frac{5}{6}\right)^4 \frac{1}{6} + \dots$ $M1$; geometric series with ratio $\frac{25}{36}$: $\frac{1/6}{1 - 25/36}$
 $M1$; $= \frac{6}{11} A1$. (b) $\frac{1}{6} \left(1 + \frac{25}{36} + \frac{625}{1296}\right) M1$; $= 0.3628 A1$.